

## \* Unit - 2 \*

### Introduction to Relational Model :-

→ Relational Model was proposed by E.F. Codd. to model data in the form of relations or tables. After designing the conceptual model of database using ER diagram, we need to convert the conceptual model in the relational model which can be implemented using any RDBMS (Relational database Management system) like SQL, MySQL etc.

### Relational model :-

→ Relational model represents how data is stored in relational databases.

→ A Relational database stores data in the form of relations (tables).

→ Consider a relation STUDENT with attributes ROLL-NO, NAME, ADDRESS, PHONE and AGE shown in table.

STUDENT

| ROLL-NO | NAME   | ADDRESS   | PHONE      | AGE |
|---------|--------|-----------|------------|-----|
| 1.      | Nishma | Hyderabad | 9455123451 | 28  |
| 2.      | Sai    | Guntur    | 9652431843 | 27  |
| 3.      | Swetha | Nellore   | 9156253131 | 26  |
| 4.      | Raji   | Ongole    | 9215635311 | 25  |

\* Attribute: Attributes are the properties that define a relation

Eg: Roll-NO, NAME

\* tuple: Each row in the relation is known as tuple.

→ The above relation contain 4 tuples.

Eg: 1 Nishma Hyderabad 94512341 28

\* Degree: The Number of attributes in the relation is known as degree of the relation.

→ The STUDENT Relation defined above has degree 5

\* column: column represents the set of values for a particular attribute.

→ The column Roll-NO is extracted from relation STUDENT

Eg: Roll-NO

↓

\* NULL values: the value which is not known or unavailable is called NULL value. it is represented by blank space.

Eg: PHONE of STUDENT having Roll-NO 4 is NULL

Concept of Domain:

→ The domain of a database is the set of all allowable values (or) attributes of the database

Eg: Gender (Male, Female, others).



Attribute: Attributes are the properties that define a relation. 2

Eg: Roll-NO, name

Tuples: Each row in the relation is known as tuple

Eg: Consider a relation employee with attributes Name, Id, Salary

employee

| Name      | Id  | Salary |
|-----------|-----|--------|
| Aishwarya | 501 | 20000  |
| Sai       | 502 | 20500  |
| Suetha    | 503 | 30000  |

In the above table

→ Aishwarya 501 20000

→ Sai 502 20500

→ Suetha 503 30000 are the tuples.

Relation:

→ A Relation is defined as a set of tuples that have same attributes. A relation consists of relation schema and relation instance.

Relation schema: A Relation schema represents the name of the relation with its attributes.

Ex: STUDENT (Roll-no, NAME, ADDRESS, PHONE and AGE) is relation schema for STUDENT

Relation instance: The set of tuples of a relation at a particular instance of time is called as relation instance.

→ STUDENT table shows the relation instance of STUDENT at a particular time. It can change whenever there is insertion, deletion or updation in the database fields (column : attributes).

→ An instance of 'employee' relation.

| tuples. | Emp-code | emp-name | Dept-name  | → Field names. |
|---------|----------|----------|------------|----------------|
|         | 01234    | John     | HR         |                |
|         | 12567    | Smith    | Sales      |                |
|         | 21678    | Sai      | Production |                |
|         | 12456    | Jay.     | design.    |                |

Fig: Instance of an 'employee' relation.

Importance of NULL values:-

→ SQL supports a special value known as NULL which is used to represent the values of attributes that may be unknown, or not apply to a tuple.

- For example, the Apartment-number attribute of an address applies only to addresses that are in apartment buildings and not to other types of residences.
- It is important to understand that a NULL value is different from zero value.
- A NULL value is used to represent a missing value, but that it usually has one of three different interpretations.
  - \* value unknown (value exists but is not known)
  - \* value not available (exists but it is purposely withheld)
  - \* Attribute not applicable (undefined for this tuple).
- It is often not possible to determine which of the meanings is intended.



## Constraints :-

→ on modeling the design of the Relational database we can put some rules (conditions) like what values are allowed to be inserted in the relation.

→ Constraints are the rules enforced on the data columns of a table. these are used to limit the type of data that can go in to a table.

→ This ensures the accuracy and reliability of the data in the database. Constraints can be either on a column level or a table level.

### Domain constraints in DBMS :-



→ In DBMS table is viewed as a combination of rows and columns.

→ For example, if you are having a column called month and you want only [Jan, Feb, March...] as values allowed to be entered for that particular column. which is referred to as domain for that particular column.

Definition: Domain constraints ensures two things.

It makes sure that the data value entered for that particular column matches with the data type defined by that column.

It shows that the constraints (NOT NULL / UNIQUE / PRIMARY KEY / FOREIGN KEY / CHECK / DEFAULT)

## Domain Constraints

| Roll-NO | Name  | Age |
|---------|-------|-----|
| 1       | Asiya | 21  |
| 2       | Bravo | 19  |
| 3       | John  | 24  |
| 4       | Max.  | 24  |

Age must be greater than 18 and age must be integer

check (Age >= 18)

→ domain constraint = data type check for the column + constraints

Example:

→ For example, we want to create a table "student" with "stu-id" field having a value greater than 100, can create a domain and table like this:

> create domain id-value int constraint id-test check (value > 100);

> create table student (stu-id id-value primary key, stu-name varchar(30), stu-age int);

key constraints in DBMS:

→ Constraints or nothing but the rules that are to be followed while entering data into columns of the database table.

→ Constraints ensure that data entered by the user into columns must be within the criteria specified by condition.

→ We have 6 types of key constraints in DBMS.

1. Not null
2. Unique
3. default
4. check
5. primary key
6. Foreign key



# 1. Not null:

→ Null represents a record where data may be missing data or data for that record may be optional.

→ Once not null is applied to a particular column, you can not enter null values to that column.

→ A not-null constraint can not be applied at table level.

Not null

NOT Null

| Roll-no | Name  | Age | phone      |
|---------|-------|-----|------------|
| 1       | Asha  | 21  | 7491901521 |
| 2       | Bravo | 19  | 8491901000 |
| 3       | John  | 24  | 9291018403 |
| 4       | max   | 24  | 7903084562 |

does not allow null values.

Null values are allowed.

## Example:

> create table student ( id int NOT null,  
name Varchar NOT null,  
Age int not null,  
Address char (25),  
salary decimal (18, 2),  
primary key (id);

→ In the above ex, we have applied not null on three columns id, name and age which means when ever a record is entered using insert statement all three columns should contain a value other than Null.

→ table have two other columns address and salary, where not null is not applied which means that you can leave the row as empty.



2. Unique: Some times we need to maintain only Unique data in the column of a database table, this is possible by using a Unique constraint.

→ Unique constraint ensures that all values in a column are Unique.

Example:

```
> create table persons ( Id int unique,  
                        last name varchar(255) not null,  
                        first name varchar(255),  
                        age int );
```

→ In the above example, as we have used unique constraint on ID column we are not supposed to enter the data that is already present, simply no two ID values are same.

Unique

Unique

| Roll-no | Name  | age | phone      |
|---------|-------|-----|------------|
| 1       | Arya  | 21  | 7491901512 |
| 2       | Barro | 19  | 8491901000 |
| 3       | John  | 24  | 929108403  |
| 4       | Max   | 24  | 7903084562 |

All values  
are different

Allow duplicate values.

3. Default: default in SQL is used to add default data to the columns.

→ When a column is specified as default with some value then all the rows will use the same value. i.e. each and every time while entering the data we need not enter that value.

→ But default column value can be customized i.e. it can be overridden when inserting a data for that row based on the requirement

### Default in DBMS

| ID-NO | name  | Company |
|-------|-------|---------|
| 1     | Asya  | abc     |
| 2     | Braro | abc     |
| 3     | John  | abc     |
| 4     | Max   | abc     |

↓  
Row with default value "abc"

#### Example:

```
> Create table emp ( id int notnull,  
                    lastname varchar(255) notnull,  
                    first name varchar(255),  
                    age int,  
                    city varchar(255) Default 'hyderabad');
```

→ As a result, when ever you insert a new row each time you need not enter a value for this default column that is entering a column value for a default column is optional.

#### 4. check:

→ check constraint ensures that the data entered by the user for that column is within the range of values or possible values specified.

#### Example:

```
> create table student ( id int,  
                        name varchar(255),  
                        age int,  
                        check (age >= 18));
```



| Roll-no | name  | age |
|---------|-------|-----|
| 1       | Ariya | 21  |
| 2       | Braro | 19  |
| 3       | John  | 24  |
| 4       | max   | 24  |

→ check.

Allow data only  
if age  $\geq 18$ .

check (Age  $\geq 18$ )

→ As we have used a check constraint as (Age  $\geq 18$ ) which means value entered by user for this age column while inserting the data must be less than or equal to 18.

### 5. primary key :

→ A primary key is a constraint in a table which uniquely identifies each row record in a database table. by enabling one or more the column in the table as primary key.

| Roll-no | Name  | age | GPA |
|---------|-------|-----|-----|
| 1       | Ariya | 21  | 4   |
| 2       | Braro | 19  | 3   |
| 3       | John  | 24  | 4.3 |
| 4       | max   | 24  | 1   |

### Creating a primary key

→ A particular column is made as a primary key column by using the primary key keyword followed with the column name.

> Create table emp ( id int,

name varchar (20),

age int,;

course varchar (10),

primary key (ID));

→ Here we have used the primary key on ID column then ID column must contain unique values i.e one ID can not be used for another student

## 6. Foreign key:

→ The Foreign key constraint is a column or list of columns which points to the primary key column of another table.

→ The main purpose of the foreign key is only those values are allowed in the present table that will match to the primary key column of another table.

Foreign key.

student\_details

| Roll-no | name  | course-id |
|---------|-------|-----------|
| 1       | Arya  | IT-101    |
| 2       | bravo | IT-102    |
| 3       | John  | IT-103    |

→ key

student\_marks

Primary  
Foreign key

| course-ID | Marks |
|-----------|-------|
| IT-101    | 100   |
| IT-102    | 99    |
| IT-103    | 93    |

primary key



### Example:

Reference table:

```
> create table customers1 ( Id int ,  
                           name varchar (20),  
                           course varchar (10),  
                           primary key (Id));
```

child table:

```
> create table customers2 ( Id int ,  
                           mark int ,  
                           references customers1 (Id));
```

### Integrity Constraints in DBMS:

→ there are two types of integrity constraints.

1. Entity Integrity constraint
2. Referential Integrity constraint

#### 1. Entity Integrity constraint:

→ Entity Integrity constraint is used to ensure the Uniqueness of each record or row in the data table.

→ Entity Integrity constraints says that no primary key can take NULL value, since using primary key we identify each tuple uniquely in a relation.

Eg:

| E Id | Name  | phone       |
|------|-------|-------------|
| 01   | Sony  | 70024 94274 |
| 02   | Pinky | 60265 26747 |
| NULL | Jalli | 92345 67892 |

## Explanation:

→ In the above relation, EID is made primary key, and the primary key can't take NULL value. but in the third tuple, the primary key is NULL, so it is violating Entity integrity constraints.

## 2. Referential Integrity Constraints:-

→ The Referential integrity constraint is specified between two relations or tables and used to maintain the consistency among the tuples in two relations.

→ This constraint is enforced through foreign key, when an attribute in the foreign key of relation  $R_1$  have the same domain as the primary key of relation  $R_2$ , then the foreign key of  $R_1$  is said to reference or refer to the primary key of relation  $R_2$ .

→ The values of the foreign key in a tuple of relation  $R_1$  can either take the values of the primary key for some tuple in relation  $R_2$ , or can take NULL values, but can't be empty.



### Example:

| ESD | Name  | DNO |
|-----|-------|-----|
| 01  | Divin | 12  |
| 02  | Dino  | 22  |
| 04  | Vipul | 14  |

Foreign key

| DNO | place  |
|-----|--------|
| 12  | Jaipur |
| 13  | Mumbai |
| 14  | Delhi  |

primary key

### Explanation:

- In the above, DNO of the first relation is the foreign key. and DNO in the second relation is the primary key.
- DNO = 22 in the foreign key of the first relation is not allowed since DNO = 22
- is not defined in the primary key of the second relation. therefore Referential integrity constraint is violated here.

## Basic SQL :-

- SQL stands for structured Query language. It is used for storing and managing data in relational database management system.
- It is standard language for relational database system. It enables a user to create, read, update and delete relational databases and tables.
- All the RDBMS like MySQL, Oracle, MS Access and SQL Server use SQL as their standard database language.
- SQL allows users to query the database in a number of ways using English like statements.

### Rules : SQL follows following rules :

- structured Query language is not case sensitive. Generally keywords of SQL are written in uppercase.
- Using the SQL statements, you can perform most of the actions in a database.
- statements of SQL are dependent on text lines. We can use a single SQL statement on one or multiple text line.

### SQL process :

- When an SQL Command is executing for any RDBMS, then the system figure out the best way to carry out the request and the SQL engine determines that how to interpret the task.



- In the process, various components are included. These components can be optimization engine, query engine, query dispatcher etc.
- All the non-SQL queries are handled by the classic query engine, but SQL query engine won't handle logical files.

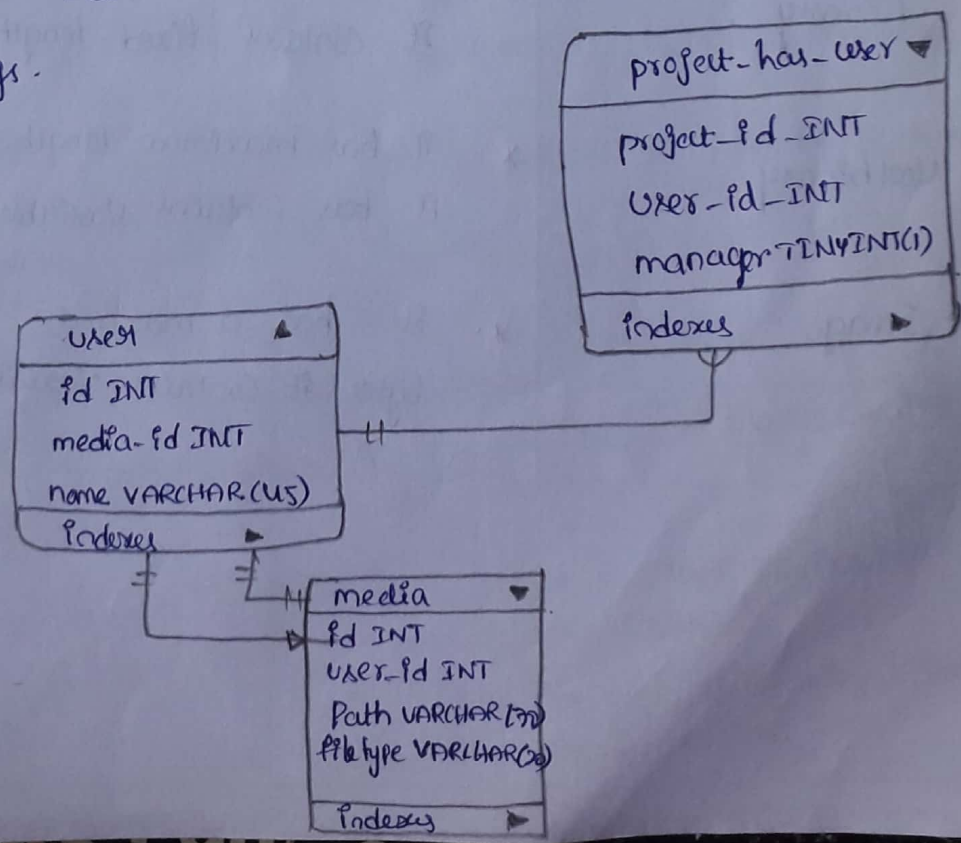
### Characteristics of SQL:

- SQL is easy to learn
- SQL is used to access data from relational database management system.
- SQL is used to describe the data
- SQL is used to create and drop the database and table
- SQL allows users to set permissions on tables, procedures, and views.

### Simple database schema:-

- A database schema is a structure that represents the logical storage of the data in a database.
- It represents the organization of data and provides information about the relationships between the tables in a given database.
- A database schema is the logical representation of a database, which shows how the data is stored logically in the entire database.
- It contains list of attributes and instruction that informs the database engine that how the data is organized and how the elements are related to each other.

- A database schema contains schema objects that may include tables, fields, packages, views, relationships, primary key, foreign key.
- In actual, the data is physically stored in files that may be in unstructured form, but to retrieve it and use it, we need to put it in a structured form. To do this a database schema is used. it provides knowledge about how the data is organized in a database and how it is associated with other data.
- A database schema object includes the following:
  - Consistent formatting for all data entries.
  - Database objects and unique keys for all data entries
  - tables with multiple columns, and each column contains it's name and datatype.
  - The given diagram is an example of a database schema it contains three tables, their data types. this also represents the relationships between the tables and primary keys as well as foreign keys.



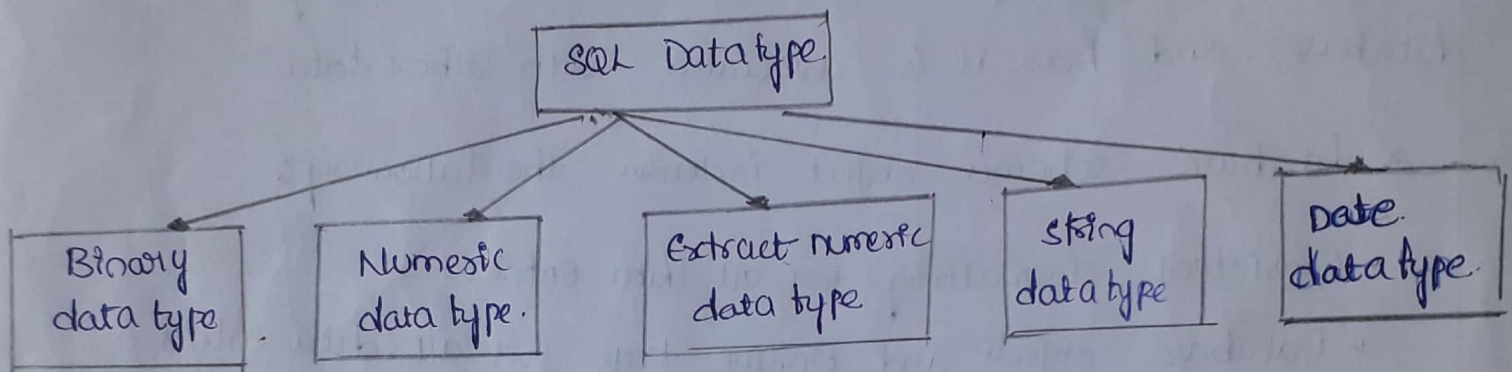


## SQL Data type :-

SQL Data type is used to define the values that a column can contain.

Every column is required to have a name and data type in the database table.

## Data types of SQL :-



### 1. Binary data types :-

These are three types of binary datatypes which are given below.

| Data type    | Description                                                                                |
|--------------|--------------------------------------------------------------------------------------------|
| * binary.    | * It has a maximum length of 8000 bytes. It contains fixed-length binary data.             |
| * varbinary. | * It has maximum length of 8000 bytes. It contains variable-length binary data.            |
| * image      | * It has a maximum length of 2,147,483,647 bytes. It contains variable length binary data. |

## 2. Numeric Data type:

| Data type | from            | to             | Description                                                        |
|-----------|-----------------|----------------|--------------------------------------------------------------------|
| * float   | -1.79E<br>+ 308 | 1.79E<br>+ 308 | * It is used to specify a floating-point value<br>eg: 6.2, 2.9 etc |
| * real    | -3.40E<br>+ 38  | 3.40E<br>+ 38  | * It specifies a single precision floating point number.           |

## 3. Exact Numeric Data type:-

| Data type   | Description                                                    |
|-------------|----------------------------------------------------------------|
| * int       | * It is used to specify an integer value.                      |
| * small int | * It is used to specify small integer value.                   |
| * bit       | * It has the number of bits to store.                          |
| * decimal   | * It specifies a numeric value that can have a decimal number. |
| * numeric   | * It is used to specify a numeric value.                       |

## 4. Date and time data types:

| Data type    | Description                                                           |
|--------------|-----------------------------------------------------------------------|
| * date       | * It is used to store the year, month, and days value.                |
| * time       | * It is used to store the hour, minute, and second values.            |
| * time stamp | * It stores the year, month, day, hour, minute, and the second value. |



## Storing datatype:

| Data type | Description                                                                                                |
|-----------|------------------------------------------------------------------------------------------------------------|
| * char    | * It has a maximum length of 8000 characters. It contains Fixed-length non-unicode characters.             |
| * VARCHAR | * It has a maximum length of 8000 characters. It contains Variable-length non-unicode characters.          |
| * Text.   | * It has a maximum length of 2,147,483,647 characters. It contains Variable length non-unicode characters. |

## Table definition: (create, alter):-

SQL table: SQL table is a collection of data which is organized in terms of rows and columns.

→ In DBMS, the table is known as relation and row as a tuple.

→ Let's see an example of the "EMPLOYEE" table.

| EMP-ID | EMP-NAME | CITY       | PHONE-NO   |
|--------|----------|------------|------------|
| 1      | Kristen  | Washington | 7289201223 |
| 2      | Anna     | Toronto    | 9378282882 |
| 3      | Jackson  | California | 9264783838 |
| 4      | Daniel   | Hawaii     | 9638482678 |

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→ In the above table, "EMPLOYEE" is the table name, "EMP-ID", "EMP-NAME", "CITY", "PHONE-NO" are the column names.

→ The combination of data of multiple columns forms a row.  
eg: 1, "Koristen", "Washington" and "7289201223" are the data of one row.

### Operations on table:

1. Create table
2. Alter table
3. Drop table

1. Create table: SQL create table is used to create a table in the database. To define the table you should define the name of the table and also define its column and column's data type.

### Syntax:

```
create table table-name ("column1" "datatype",  
                          "column2" "datatype",  
                          "column3" "datatype",  
                          :  
                          "column N" "datatype");
```

### Example:

```
SQL > create table employee ( emp_id int,  
                               emp_name varchar(25),  
                               phone_no int,  
                               Address. char(30));
```

→ If you create the table successfully, you can verify the table by looking at the message by the SQL server. else you can use DESC command as follows:



SQL > Desc employee;

| Field    | type        | NULL    | Default | Extra    |
|----------|-------------|---------|---------|----------|
| emp_id   | int(11)     | NO      | NULL    |          |
| emp-name | varchar(25) | NO      | NULL    |          |
| phone-no | NO          | int(11) | NULL    |          |
| Address  | YES         |         | NULL    | char(30) |

### Alter table:

→ The alter table command adds, delete, or modifies columns in a table.

→ The alter table command also adds and deletes various constraints in a table.

→ The following SQL adds an "Email" column to the "employee" table:

### Syntax:

Alter table table name add column1 datatype;

### Example:

Alter table employee add email varchar(255);

SQL > Desc employee;

| Field    | type         | NULL    | default | extra.   |
|----------|--------------|---------|---------|----------|
| emp_id   | int(11)      | NO      | NULL    |          |
| emp-name | varchar(25)  | NO      | NULL    |          |
| phone-no | NO           | int(11) | NULL    |          |
| Address  | YES          |         | NULL    | char(30) |
| email    | varchar(255) |         | NULL    |          |

### 3. Drop table:

- The drop table command deletes a table in the database.
- The following example SQL deletes the table "employee"

Syntax: drop table tablename;

Example: drop table employee;

- Dropping a table results in loss of all information stored in the table.

### Different DML operations: (insert, delete, update)

- DML - Data Manipulation ~~Commands~~ Language.
- Data Manipulation commands are used to manipulate data in the database.
- Some of the data Manipulation Commands are.

1. insert
2. update
3. delete

1. insert: SQL insert statement is a SQL query. It is used to insert a single or multiple records in a table.

Syntax: insert into table name values (value 1, value 2, value 3);

let's take an example of table which has 3 records within it.

- > insert into student values ('Alekhya', 501, 'Hyderabad');
- > insert into student values ('Deepthi', 502, 'Guntur');
- > insert into student values ('Ramya', 503, 'Nellore');



The following table will be as follow:

| Name    | Id  | city.     |
|---------|-----|-----------|
| Alekhya | 501 | Hyderabad |
| Deepthi | 502 | Guntur    |
| Ramya   | 503 | Nellore   |

## 2. update:

→ The SQL commands update are used to modify the data that is already in the database. ~~the SQL~~

→ SQL update statement is used to change the data of the records held by tables. which row is to be update, it is decided by condition. to specify condition, we use WHERE clause.

→ The update statement can be written in following form:

Syntax: update table\_name set column\_name = expression  
where condition;

Example: let's take an example : here we are going to update an entry in the table.

> update students set name = 'Ravi' where id = 503;

→ After update the table.

| Name    | id  | city       |
|---------|-----|------------|
| Alekhya | 501 | Hyderabad. |
| Deepthi | 502 | Guntur     |
| Ravi    | 503 | Nellore    |

### 3. delete :-

→ The SQL delete statement is used to delete rows from a table. Generally delete statement removes one or more records from a table.

#### Syntax:-

→ Delete from table-name [Where condition];

#### Example:-

→ Let us take a table, named "student" table.

> Delete from students where id = 501;

→ Resulting table after the query:

| name    | id  | city.   |
|---------|-----|---------|
| Deepthi | 502 | Guntur  |
| Ravi    | 503 | Nellore |

## UNIT II

Relational Model: Introduction to relational model, concepts of domain, attribute, tuple, relation, importance of null values, constraints (Domain, Key constraints, integrity constraints) and their importance, Relational Algebra, Relational Calculus. BASIC SQL: Simple Database schema, data types, table definitions (create, alter), different DML operations (insert, delete, update).